

Input protection:

Each of the big HV parts are crucial to the input protection. Especially, R8 limits the current and holds off the brunt of the voltage. R16 limits the downstream voltage to 12kV instead of ~80V. So during an 80V transient, R8 will drop ~74V (and limit the max current to ~7A), and R16 will drop ~14V.

R8 must be able to withstand >> 80V transients without arcing, shorting, or burning, though fusing open is fine.

C1 sets the maximum voltage we can measure (minimum varistor voltage), and what the 3.3M string has to be able to tolerate (maximum standoff voltage).

R17 is just there to protect C1. If we're in AC mode during a transient, to keep it from seeing any big V_{dc} voltage.

This forms an inverting voltage attenuator (amplifier with gain = -1).
 U2 – US switch which range is active.
 R20 – R22 set the attenuation ratio (ratio relative to R19+R22+R25).
 The capacitors in parallel with gain resistors are for AC compensation (to get a decent high frequency response). When no switches are closed, R28 sets the gain to 3:1, this becomes unstable much below that.
 R19, R23, R29 must have a fairly high withstand voltage. At 1A transients they may see >> 14V. Same for the 330Ω caps and R16.
 R10 is there to add some roll off at higher frequencies.

Voltage ranges are 0 indexed, but, as closed switches makes range 0 active

Ideally 1n=89p & 10n=890p protty 150p is 1n fine...

Vos increases to ~3mV at Vcc=1V

Current Shunts

=>Diodes MMT100A 1000uA @ 40V (what I've been using!)
 BSS55A - 2-7 uAmps 1000uA @ 25V max
 So's good=ones BSS58A

High Side > Low Side Shunt Voltage Translator
 (explanation text below)

-5V rail

High Side Opamp Voltage Regulator

This is a set of switchable current shunts, to measure current going to the DUT. The voltage (V_{in}) is the high -5V rail and bottom of the shunt(s) this sent to the High > Low side translator, and will eventually go to the ADC.

Q1, Q4, Q18 are low-leakage pFets, normally all but one is open circuit, closed one sets the active range.

Q1, Q3, ... Q17 are just to let the MCU turn on and off those pFets.

R6, R15, ... R37 are the current shunts.

MM524V2715 Bias 1?
 T0D5, 10K value? Bias 1?
 Green path=high
 MM524V01 might be better?
 MM524V91-13-08 from diyark looks good

Positively TUNEDCROCKE If this doesn't work
 10000 double check that high-low buffer doesn't saturate

Q11 is present to limit negative voltages.
 If the negative voltage is high by the current source, and Q11 turns on, and has no effect. If negative voltage is fed into the jack, source will be pulled toward 0V, turning Q11 off and protecting the downstream circuits.

The diagram shows a dual inverter driver circuit for a brushless motor. It consists of two MOSFETs (Q1, Q2) driven by a microcontroller (MCU) through gate drivers (D1, D2). The MOSFETs are connected to a motor winding. The circuit also includes a feedback loop with a resistor network (R1, R2, R3, R4, R5, R6, R7, R8, R9, R10, R11, R12, R13, R14, R15, R16, R17, R18, R19, R20, R21, R22, R23, R24, R25, R26, R27, R28, R29, R30, R31, R32, R33, R34, R35, R36, R37, R38, R39, R40, R41, R42, R43, R44, R45, R46, R47, R48, R49, R50, R51, R52, R53, R54, R55, R56, R57, R58, R59, R60, R61, R62, R63, R64, R65, R66, R67, R68, R69, R70, R71, R72, R73, R74, R75, R76, R77, R78, R79, R80, R81, R82, R83, R84, R85, R86, R87, R88, R89, R90, R91, R92, R93, R94, R95, R96, R97, R98, R99, R100) and a feedback loop with a resistor network (R1, R2, R3, R4, R5, R6, R7, R8, R9, R10, R11, R12, R13, R14, R15, R16, R17, R18, R19, R20, R21, R22, R23, R24, R25, R26, R27, R28, R29, R30, R31, R32, R33, R34, R35, R36, R37, R38, R39, R40, R41, R42, R43, R44, R45, R46, R47, R48, R49, R50, R51, R52, R53, R54, R55, R56, R57, R58, R59, R60, R61, R62, R63, R64, R65, R66, R67, R68, R69, R70, R71, R72, R73, R74, R75, R76, R77, R78, R79, R80, R81, R82, R83, R84, R85, R86, R87, R88, R89, R90, R91, R92, R93, R94, R95, R96, R97, R98, R99, R100). The circuit is powered by a 5V supply and a 12V supply. The motor is connected to the output of the inverter. The circuit is labeled with component values and pin numbers. The circuit is a dual inverter driver for a brushless motor.

mV is just a straight 1:1 Buffer into the ADC, for maximum sensitivity, and low Z.

Everything but the opamp itself and C5 (simple low pass filter to reduce noise) is input protection.

The 1M resistor is there to current limit overvoltages, so nothing blows out. Its value should be non-critical.

The diagram illustrates a power management circuit for a Raspberry Pi 4. It starts with a 9V battery input connected to a reverse voltage protection diode (D1). The battery is also connected to a 5V regulator (U1) which provides a 3.3V output. A power switch (P1) is controlled by a push button (Per_Button) and a 5V regulator (U3) which provides a 3.3V output. The circuit is populated with various components like resistors, capacitors, and diodes, and is labeled with component values and pin numbers.

Power Switch
When SW1 is closed, it turns on U3B3, which pulls the P_FET gate to ground. This turns it on and powers everything. As soon as the uC boots it turns on the BJT, opening P_FET and poweroff turns off BJT, which turns off P_FET, cutting power until rebooted.

TQD00: Test fb loop in hardware. sim says OK but probably improvable

TODD: check the gnd pad here

Route as cuttable jumper

Pin-compatible with a 24 bit version (MCP3461), though it only has ~19 effective bits typically. The voltage reference is optional for better stability, it hasn't been characterized for the MCP3408's stability so may be completely redundant.

Since the output runs at 5V, might be possible for it to saturate >5V into the ADC, which would be a problem. The trace around R108 can be cut, and R108 populated to alternate bits (limits current to what the steering diodes can handle).

The 220R on the left and the MUX steering diodes "should" prevent that from being a problem though.

MCP3402 requires a pullup on IRQ to function normally!

The 10K resistor loads the filter bit (capacitance multiplier) so there's not a huge swing in voltage drop. The 1u cap is decoupling, again for filtering. The two resistors are a voltage divider so the MCU can read this voltage out during operation, to make sure we've got sufficient voltage output for the DUT, headroom, and drop.

The purpose of this block is to measure the voltage developed across the shunts (without altering it), and to convert this voltage to the -40V rail, to relative to COM, so the ADC can measure it. US24, US26, and 843 together act as a voltage controlled current source (with negative control voltage, and that source current). This means the current going out of Q24 will be proportional to the voltage (and thus current) across the shunt, so we can use this to measure the current through the shunt. So4 then converts that current back into a voltage, relative to COM, which is what we want. Upshot, voltage coming out of Q24 (V24) is proportional to the current through the DUT (ratio depending on the shunt value), so we can do resistance calculations. To help, voltage input from going all the way to the rail (Spamp) is "r-r", but sometimes there's funky behavior at the rail, so the -15V drop from the filter is used. So the opamp supply is the "unfiltered" side (-V), and the noninverting input is the filtered side, -40V. This does mean we rely on the power supply rejection ratio of the opamp to filter supply noise, but that should be fine.

[illegible][illegible]